Multimedia Computing

By

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Content

- Background
- Patterns and pattern classes
- Pattern classification by prototype matching
- Template matching
- Statistical classifier
- additional details

Pattern classification

- A pattern is an arrangement of descriptors,
- The name feature is used often in the pattern recognition literature to denote a descriptor.
- A pattern class is a family of patterns that share some common properties

An Example

- "Sorting incoming Fish on a conveyor according to species using optical sensing"

Species

Sea bass

Salmon

Problem Analysis

- Set up a camera and take some sample images to extract features

  - Length
  - Lightness
  - Width
  - Number and shape of fins
  - Position of the mouth, etc…

  - This is the set of all suggested features to explore for use in our classifier!

Preprocessing

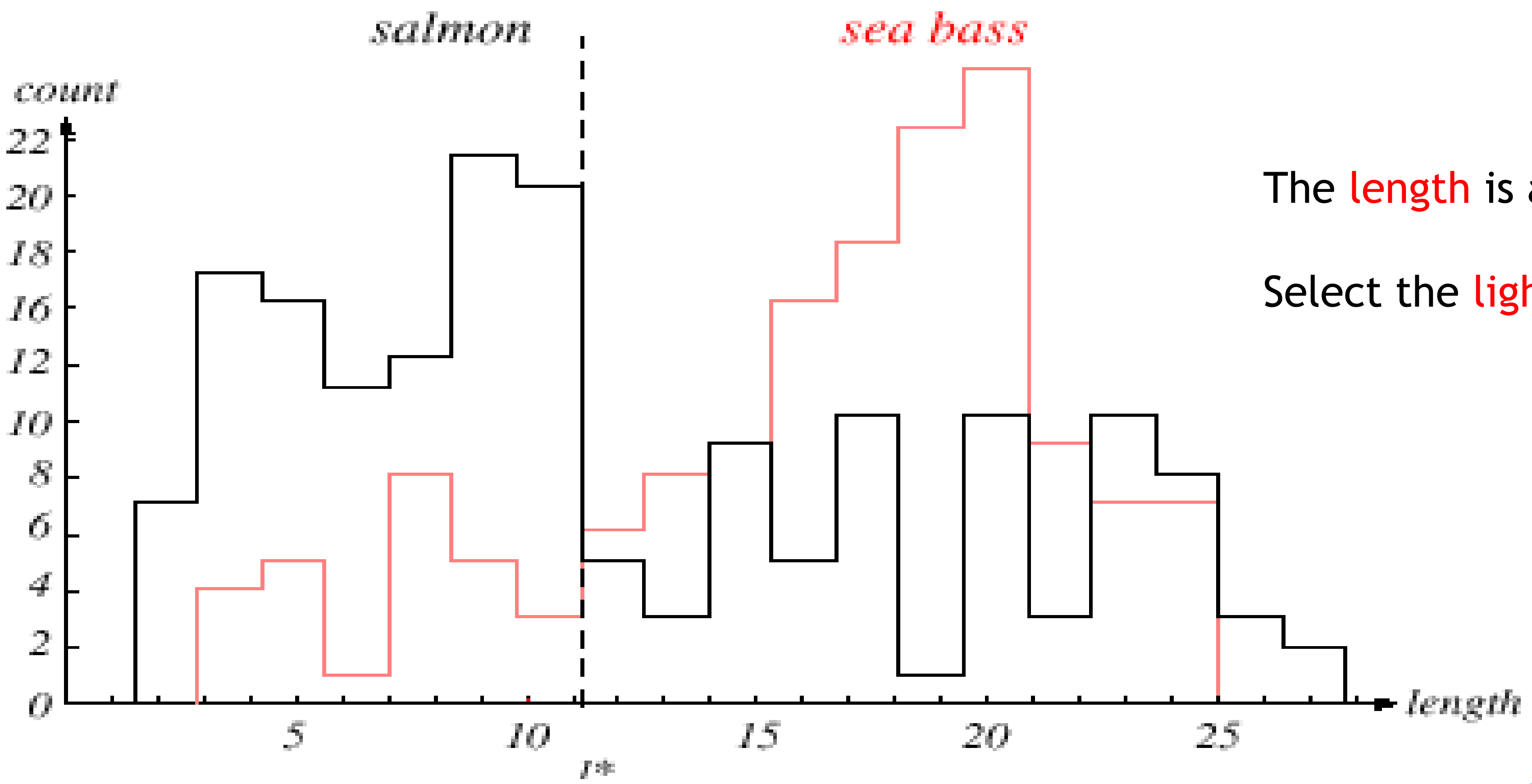
- Use a segmentation operation to isolate fishes from one another and from the background
- Information from a single fish is sent to a feature extractor whose purpose is to reduce the data by measuring certain features
- The features are passed to a classifier

Preprocessing

Classification

- Select the length of the fish as a possible feature for discrimination

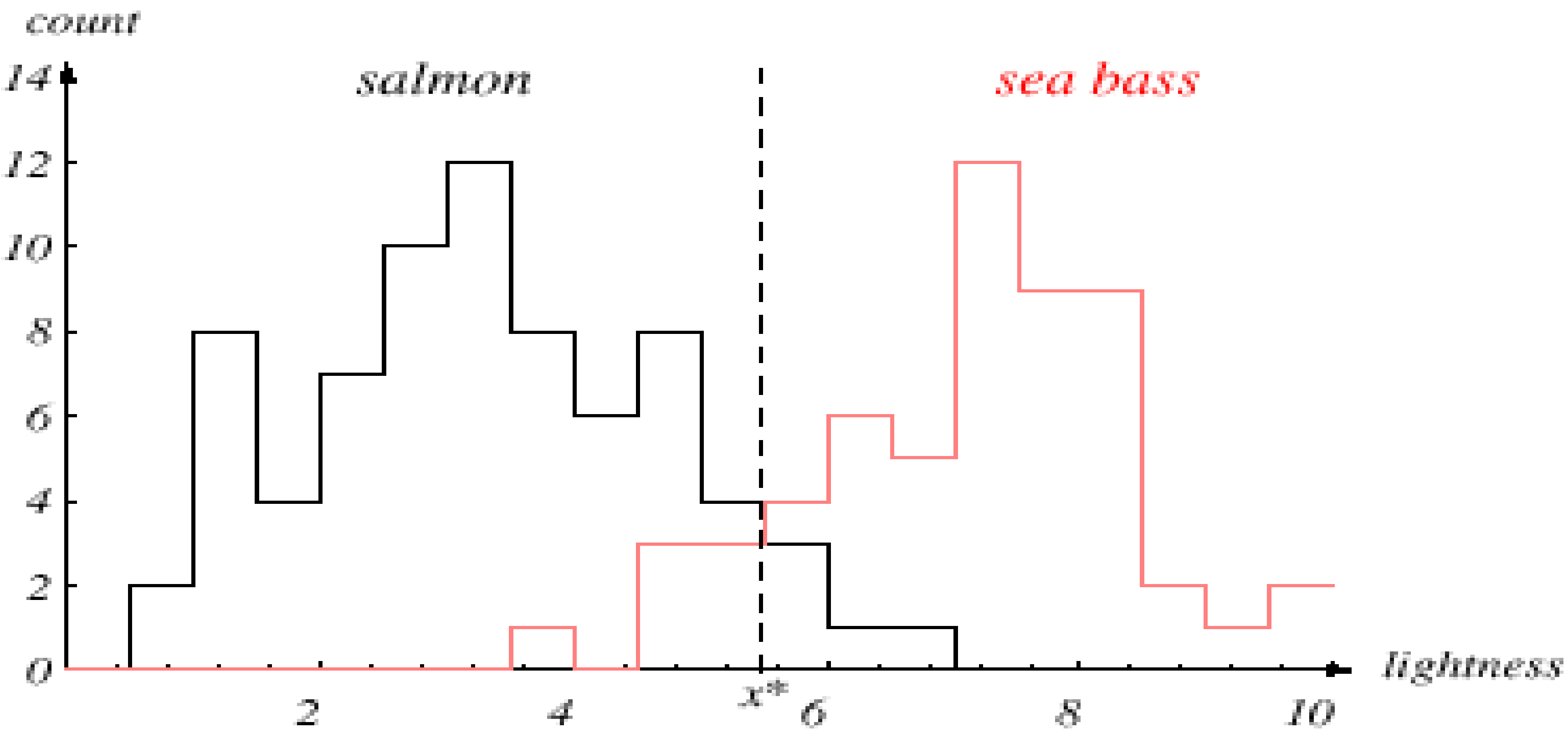
Feature identification



The length is a poor feature alone!

Select the lightness as a possible feature.

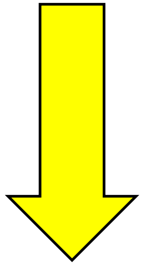
Feature identification



Threshold decision boundary and cost relationship

- Move our decision boundary toward smaller values of lightness in order to minimize the cost (reduce the number of sea bass that are classified salmon!)

Task of decision theory

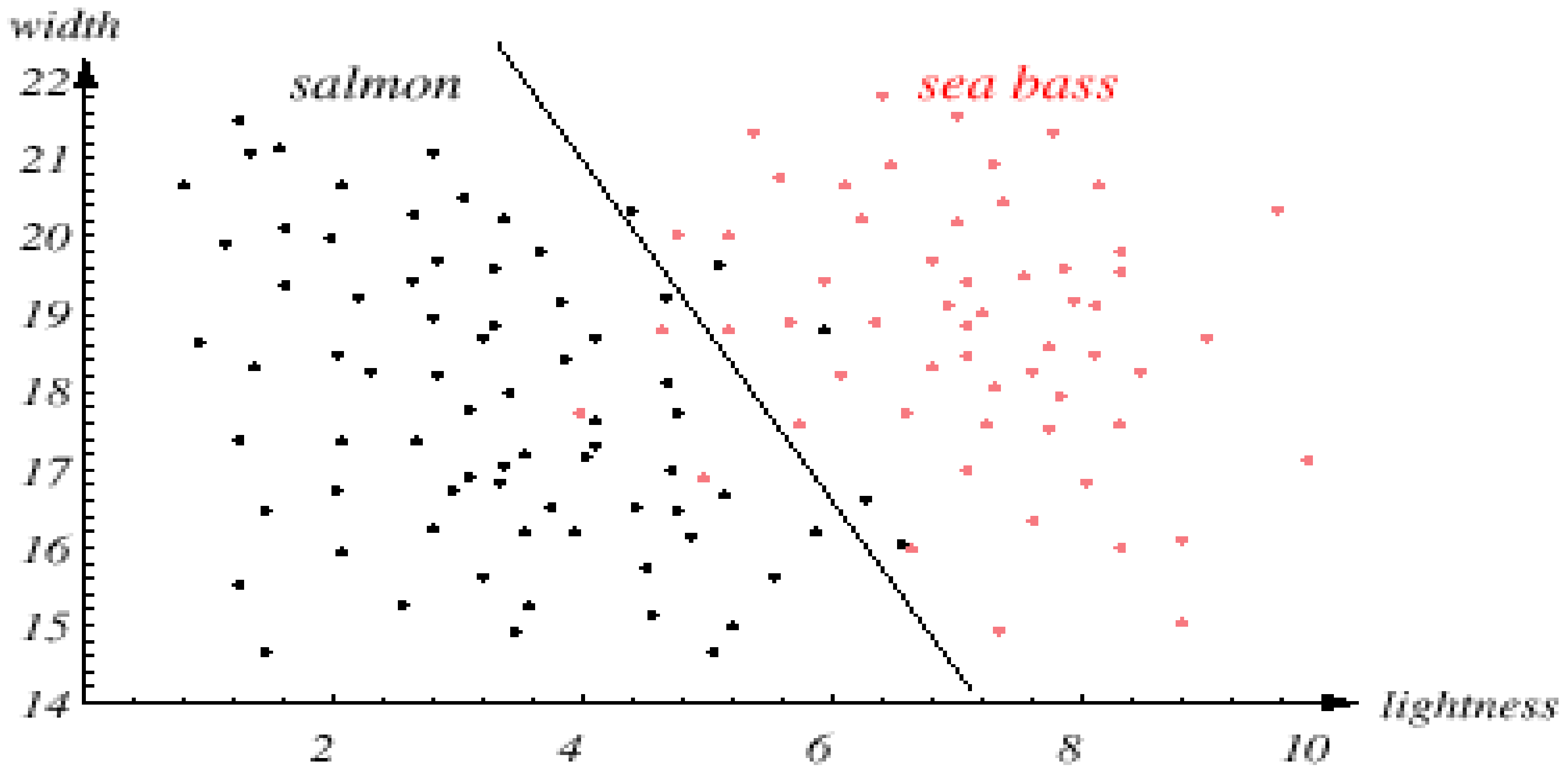


Adopt the lightness and add the width of the fish

Fish $\Rightarrow$ $x^T = [x_1, x_2]$

Lightness **Width**

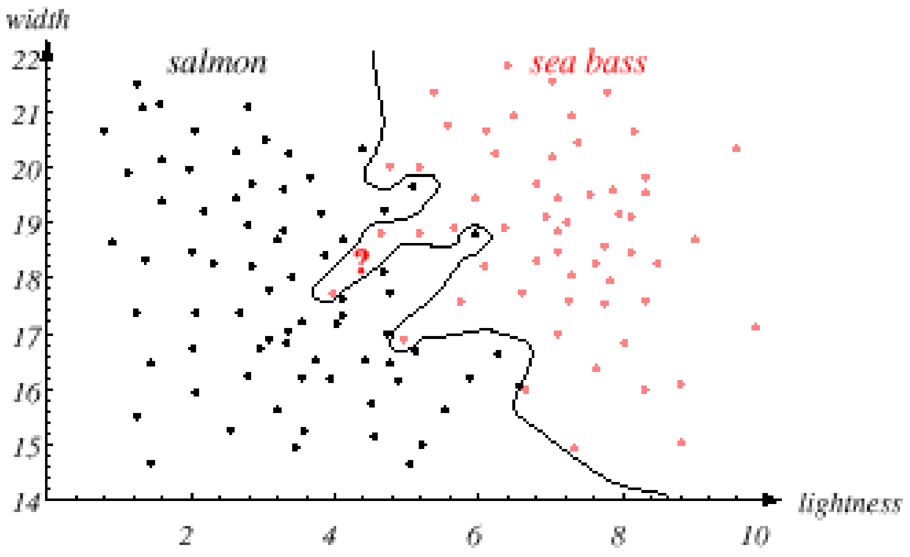
Threshold decision boundary and cost relationship



Best decision boundary

- We might add other features that are not correlated with the ones we already have. A precaution should be taken not to reduce the performance by adding such “noisy features”

- Ideally, the best decision boundary should be the one which provides an optimal performance such as in the following figure:

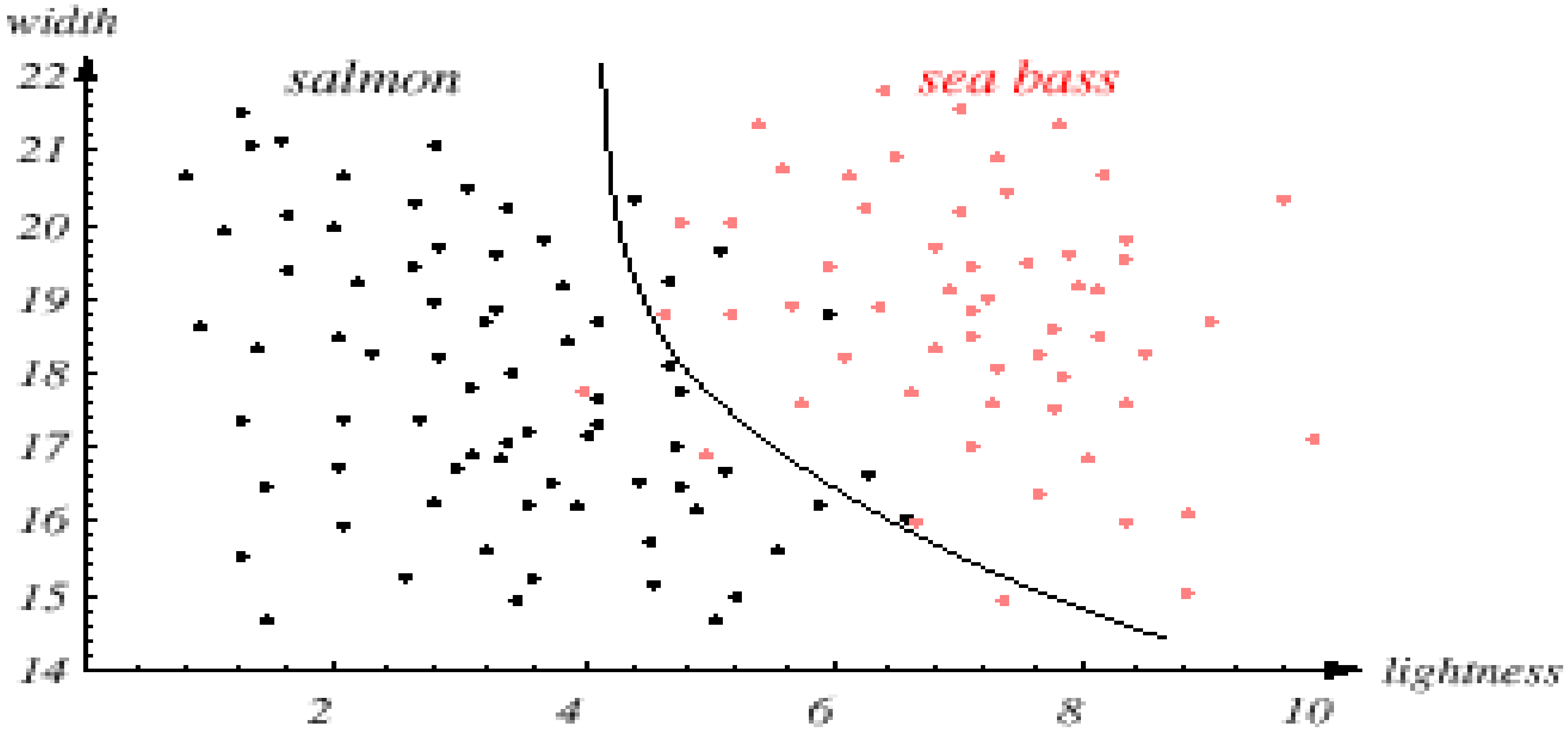


Classify Input

- However, our satisfaction is premature because the central aim of designing a classifier is to correctly classify novel input

Issue of generalization!

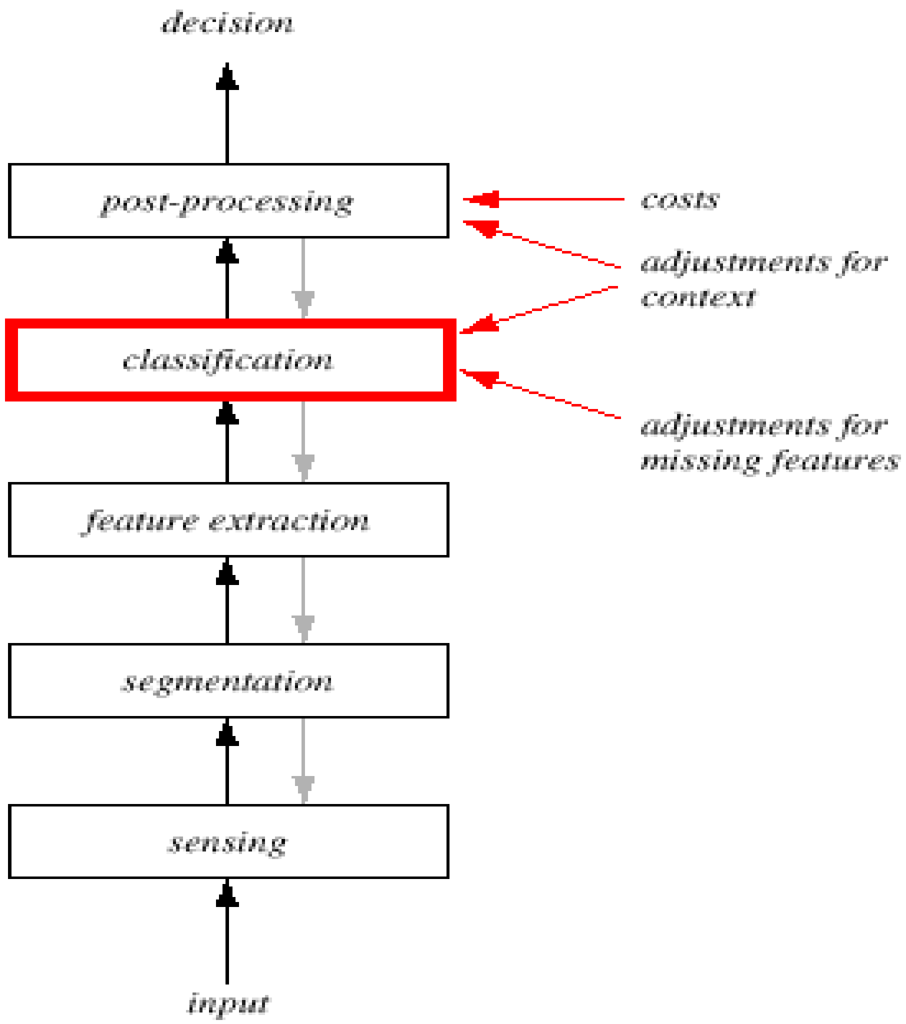
Classified Input



Pattern Recognition Systems

- Sensing
  - Use of a transducer (camera or microphone)
  - PR system depends of the bandwidth, the resolution sensitivity distortion of the transducer
- Segmentation and grouping
  - Patterns should be well separated and should not overlap

Pattern Recognition system

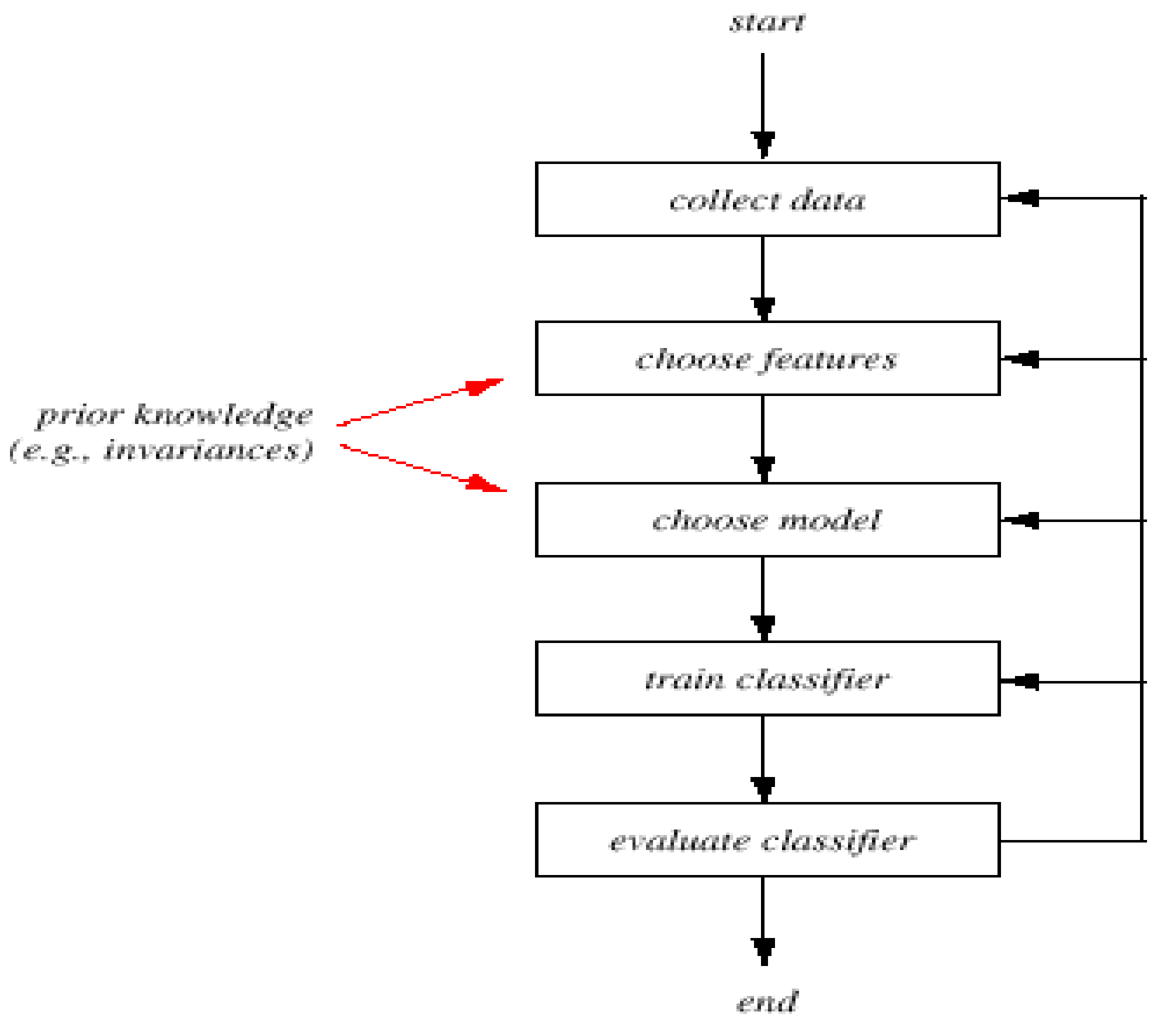


- Feature extraction
  - Discriminative features
  - Invariant features with respect to translation, rotation and scale.
- Classification
  - Use a feature vector provided by a feature extractor to assign the object to a category
- Post Processing
  - Exploit context input dependent information other than from the target pattern itself to improve performance

The Design Cycle

- Data collection
- Feature Choice
- Model Choice
- Training
- Evaluation
- Computational Complexity

The Design Cycle



Data Collection

- How do we know when we have collected an adequately large and representative set of examples for training and testing the system?

Feature Choice

- Depends on the characteristics of the problem domain. Simple to extract, invariant to irrelevant transformation insensitive to noise.

Model Choice

- Unsatisfied with the performance of our fish classifier and want to jump to another class of model

Training

- Use data to determine the classifier. Many different procedures for training classifiers and choosing models

Evaluation

- Measure the error rate (or performance and switch from one set of features to another one

Computational Complexity

- What is the trade-off between computational ease and performance?
- How an algorithm scales as a function of the number of features, patterns or categories?

Learning and Adaptation

- Supervised learning
  - A teacher provides a category label or cost for each pattern in the training set
- Unsupervised learning
  - The system forms clusters or “natural groupings” of the input patterns

Statistical Pattern Recognition

- The data is reduced to vectors of numbers and statistical techniques are used for the tasks to be performed

Structural Pattern Recognition

- The data is converted to a discrete structure (such as a grammar or a graph) and the techniques are related to computer science subjects (such as parsing and graph matching).

Classification in Statistical Pattern Recognition

- A class is a set of objects having some important properties in common
- A feature extractor is a program that inputs the data (image) and extracts features that can be used in classification.
- A classifier is a program that inputs the feature vector and assigns it to one of a set of designated classes or to the “reject” class.

Feature Vector Representation

- X=[x1, x2, … , xn], each xj a real number
- xj may be an object measurement
- xj may be count of object parts
- Example: object rep. [#holes, #strokes, moments, …]

```
00000000010000000000
00000000110000000000
00000000101000000000
00000001000010000000
00000010000010000000
00000100000001000000
00001000000000100000
00001100111111110000
00001111110000010000
00011000000000011000
00010000000000001100
00110000000000000100
00110000000000000110
00100000000000000010
00100000000000000010
01100000000000000010
01000000000000000000
00000000000000000000
```

```
00000000011110000000
00000001100001100000
00000011000000110000
00001100000000011000
00001000000000001000
00001100000000010000
00000111000000100000
00000011100111100000
00000000111100000000
00000011000111000000
00001100000000110000
00011000000000011000
00110000000000001000
00100000000000001100
00010000000000011000
00011000000000010000
00001000000000110000
00000011111110000000
```

Character set

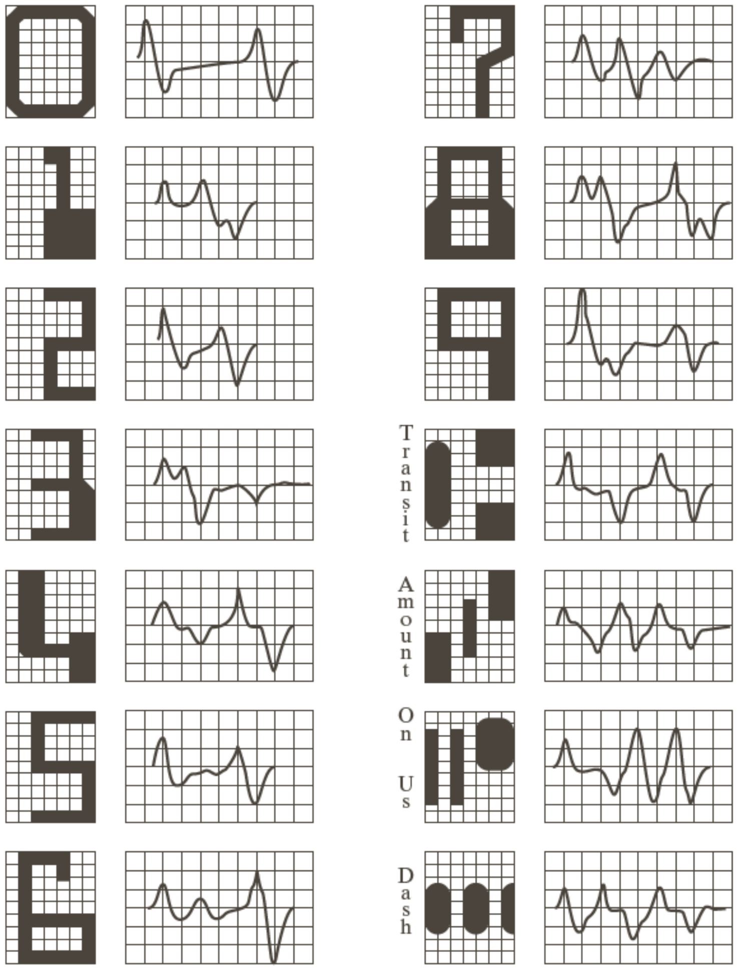


FIGURE 12.7 American Bankers Association E-13B font character set and corresponding waveforms.

Possible features for character recognition

(class) character	area	height	width	number #holes	number #strokes	(cx,cy) center	best axis	least inertia
'A'	medium	high	3/4	1	3	1/2,2/3	90	medium
'B'	medium	high	3/4	2	1	1/3,1/2	90	large
'8'	medium	high	2/3	2	0	1/2,1/2	90	medium
'0'	medium	high	2/3	1	0	1/2,1/2	90	large
'1'	low	high	1/4	0	1	1/2,1/2	90	low
'W'	high	high	1	0	4	1/2,2/3	90	large
'X'	high	high	3/4	0	2	1/2,1/2	?	large
'*'	medium	low	1/2	0	0	1/2,1/2	?	large
'-'	low	low	2/3	0	1	1/2,1/2	0	low
'/'	low	high	2/3	0	1	1/2,1/2	60	low

Template matching

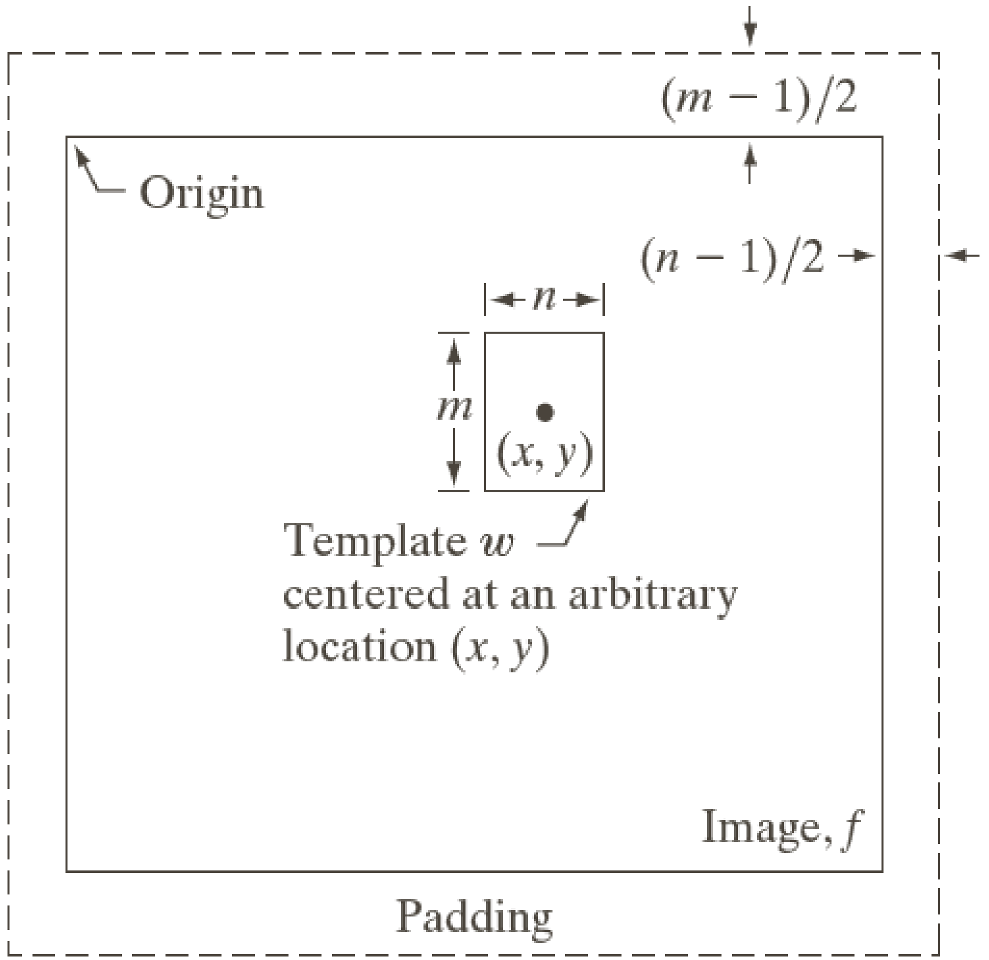


FIGURE 12.8
The mechanics of template matching.

Few terminology

- Classes: set of m known categories of objects

  (a) might have a known description for each

  (b) might have a set of samples for each
- Reject Class:

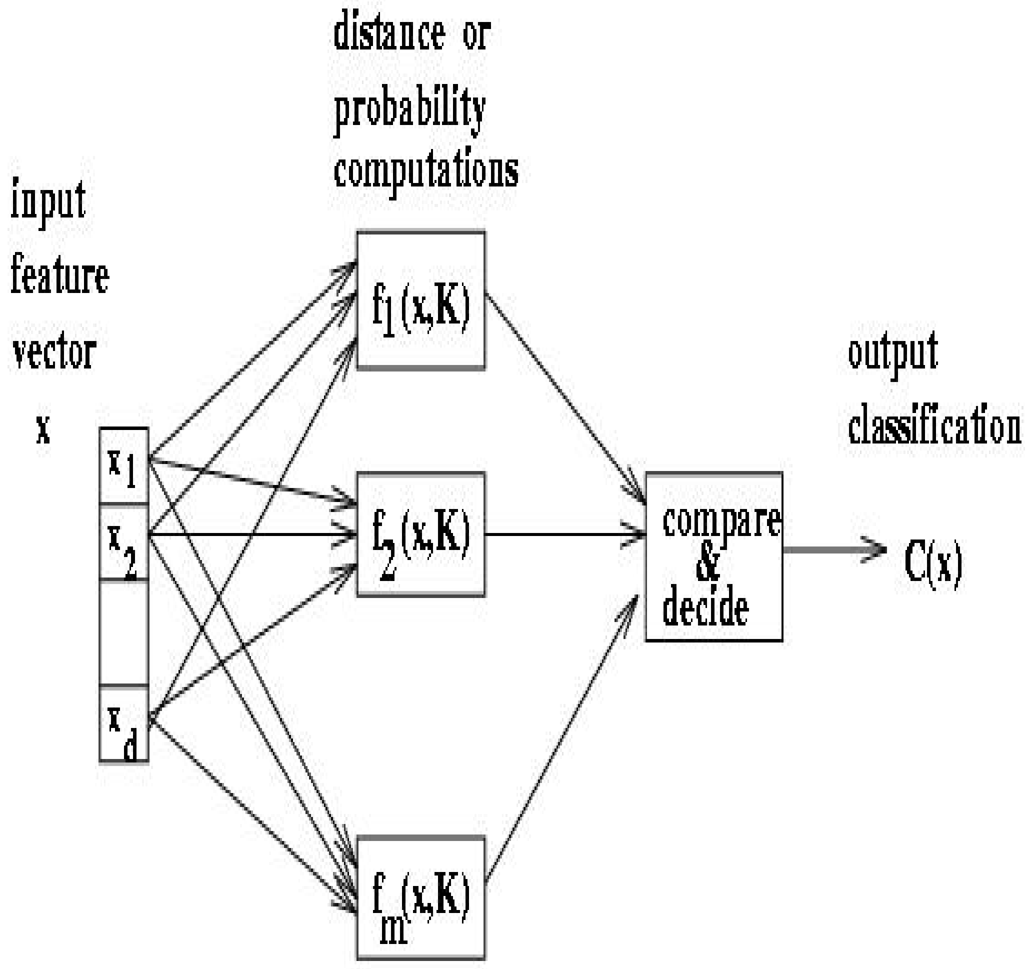
  a generic class for objects not in any of

  the designated known classes
- Classifier:

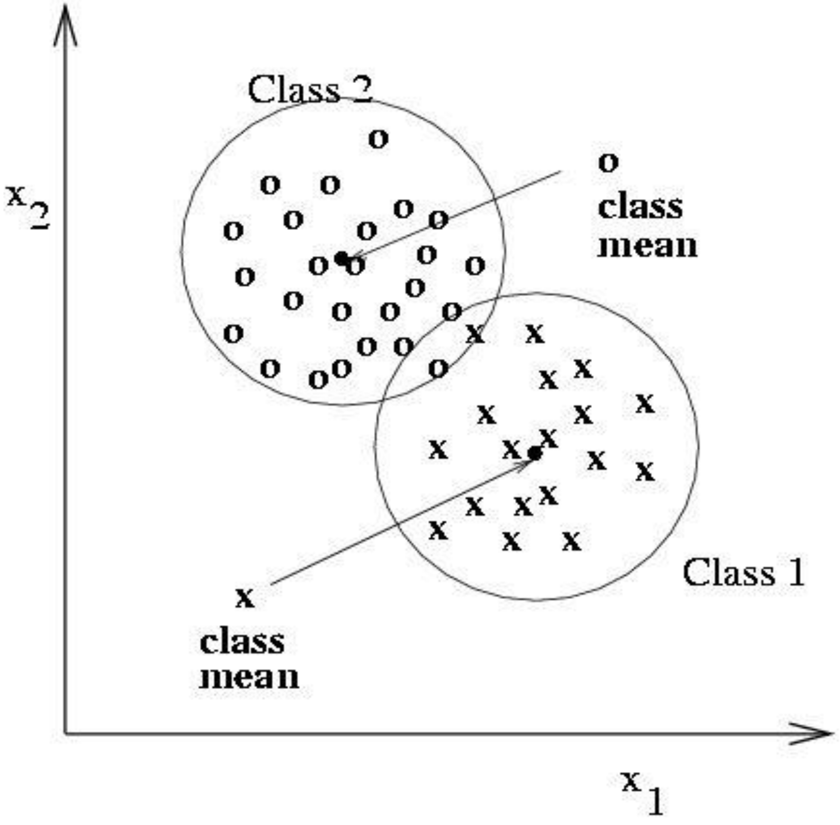
  Assigns object to a class based on features

Discriminant functions

- Functions f(x, K) perform some computation on feature vector x
- Knowledge K from training or programming is used
- Final stage determines class

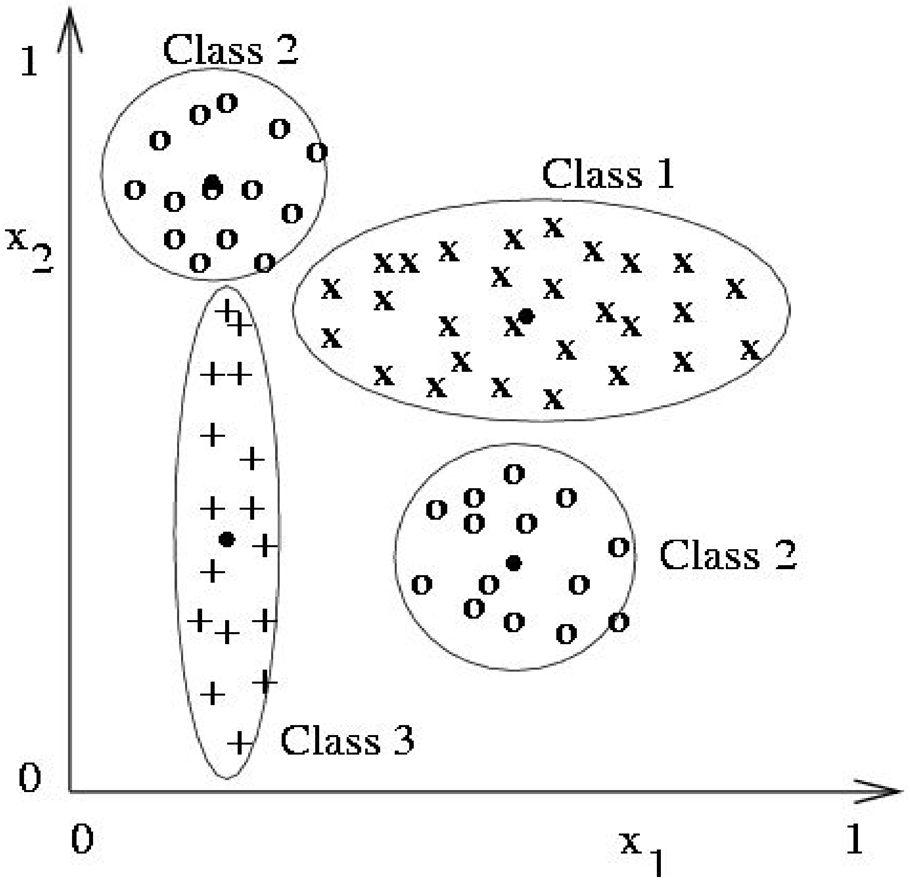


Classification using nearest class mean



- Compute the Euclidean distance between feature vector X and the mean of each class.
- Choose closest class, if close enough (reject otherwise)

Nearest mean might yield poor results with complex structure



- Class 2 has two modes; where is its mean.
- But if modes are detected, two subclass mean vectors can be used

Scaling coordinates by std dev

We can compute a modified distance
from feature vector $\mathbf{x}$ **to** class mean vector $\mathbf{x_c}$
by scaling by the spread, or *standard deviation*,
σ_i of class c along each dimension i.

scaled Euclidean distance from $\mathbf{x}$ to class mean $\mathbf{x_c}$:

$$\| \mathbf{x} - \mathbf{x_c} \| = \sqrt{\Sigma_{i=1,d}((\mathbf{x}[i] - \mathbf{x_c}[i])/\sigma_i)^2}$$

In the previous 3 class problem, an observed X near the top of the Class 3 distribution will scale to be closer to the mean of Class 3 than to the mean of Class 2. Without scaling, X would be closer to the mean of Class 2.

Nearest Neighbor Classification

- Keep all the training samples in some efficient look-up structure.
- Find the nearest neighbor of the feature vector to be classified and assign the class of the neighbor.
- Can be extended to K nearest neighbors.

Bayesian decision-making

- classify into class ω_i that is most likely based on observations $\mathbf{X}$
- In order to compute the likelihoods given the measurement $\mathbf{X}$, the following distributions are needed.

$$\begin{aligned}
\textbf{class conditional distribution}: &\quad p(x|\omega_i)\ for\ each\ class\ \ omega_i &(1)\\
\textbf{a priori probability}: &\quad P(\omega_i)\ for\ each\ class\ \omega_i &(2)\\
\textbf{unconditional distribution}: &\quad p(x) &(3)
\end{aligned}$$

- use Bayes rule if all of the classes ω_i are disjoint

$$P(\omega_i|x) = \frac{p(x|\omega_i)P(\omega_i)}{p(x)} = \frac{p(x|\omega_i)P(\omega_i)}{\Sigma_{i=1,m}\, p(x|\omega_i)P(\omega_i)} \qquad (4)$$

Flower classification

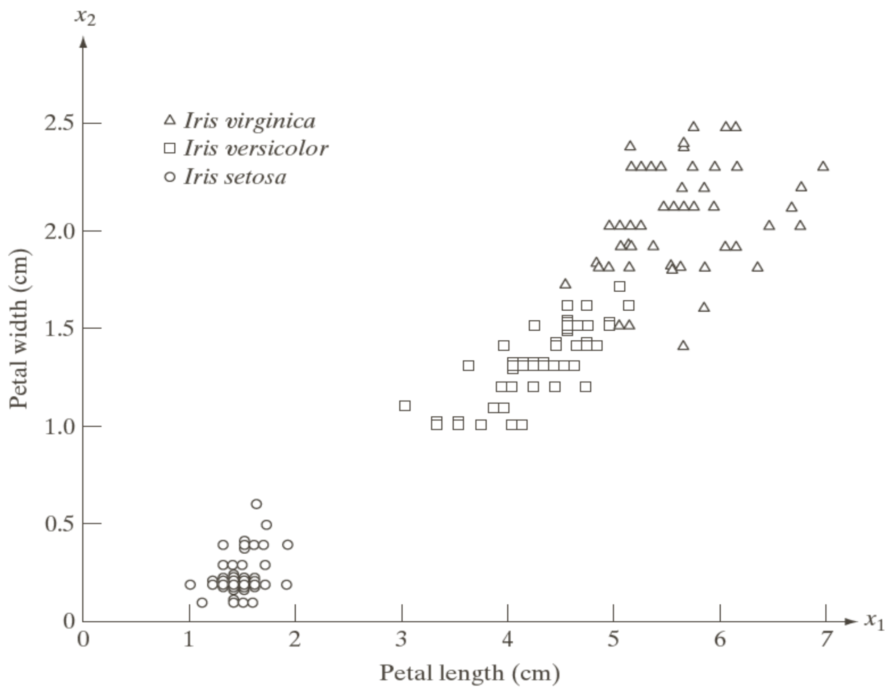


FIGURE 12.1
Three types of iris flowers described by two measurements.

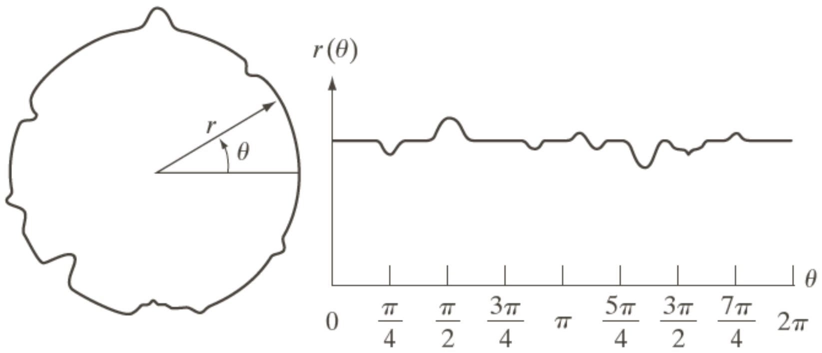
Iris flower dataset

Bayes classification of multispectral image data

	Training Patterns					Independent Patterns					
		Classified into Class						Classified into Class			
Class	No. of Samples	1	2	3	% Correct	Class	No. of Samples	1	2	3	% Correct
1	484	482	2	0	99.6	1	483	478	3	2	98.9
2	933	0	885	48	94.9	2	932	0	880	52	94.4
3	483	0	19	464	96.1	3	482	0	16	466	96.7

TABLE 12.1
Bayes classification of multispectral image data.

Noisy object



a b

FIGURE 12.2 A noisy object and its corresponding signature.

Noisy object

FIGURE 12.4 Satellite image of a heavily built downtown area (Washington, D.C.) and surrounding residential areas. (Courtesy of NASA.)

Flower classification

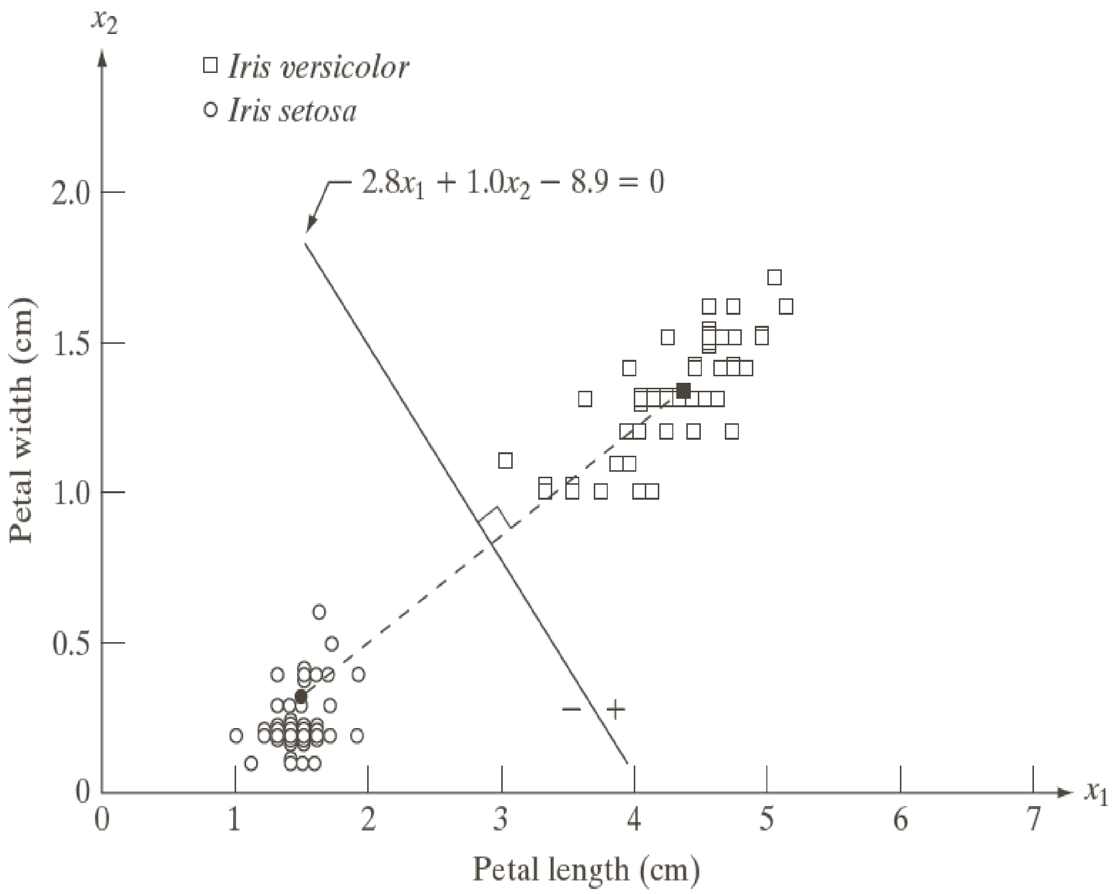
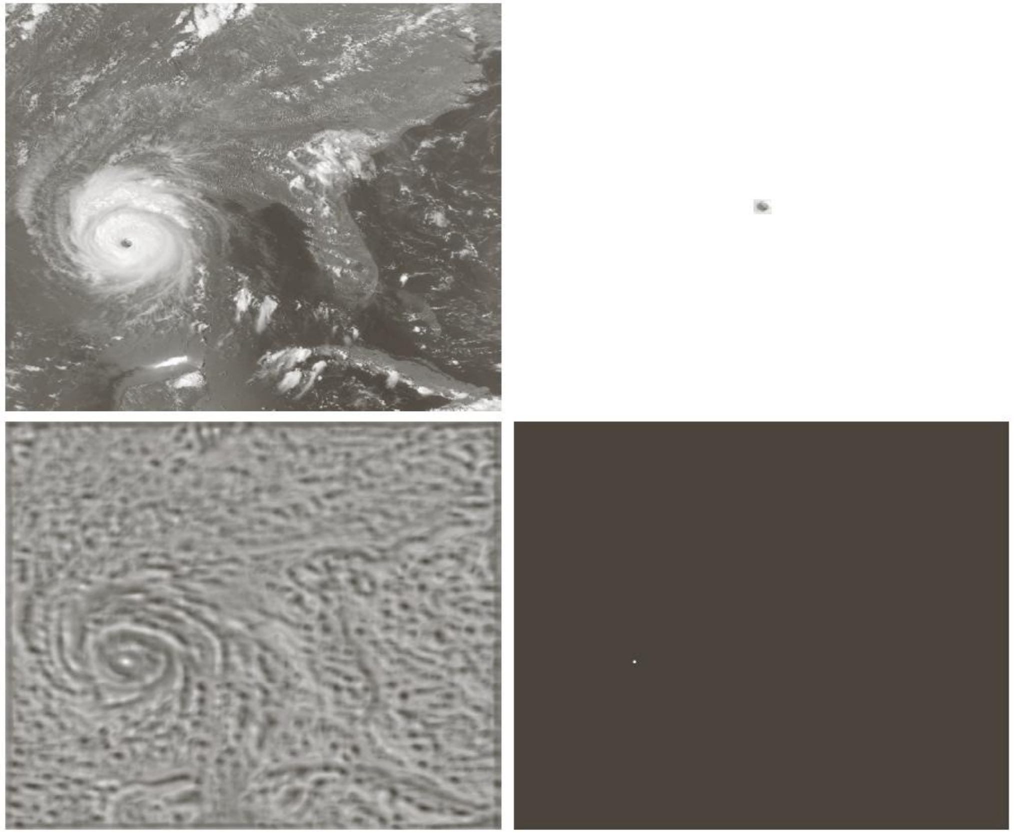


FIGURE 12.6 Decision boundary of minimum distance classifier for the classes of *Iris versicolor* and *Iris setosa*. The dark dot and square are the means.

Object recognition

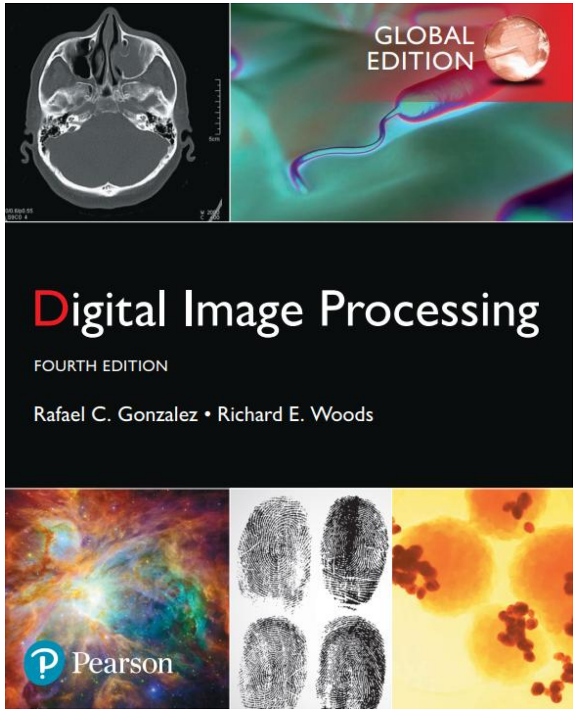


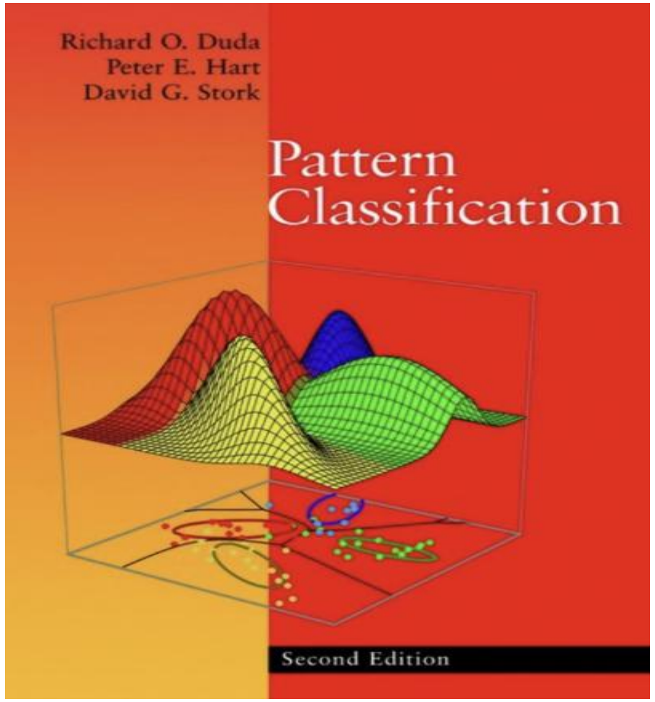
a b
c d

FIGURE 12.9
(a) Satellite image of Hurricane Andrew, taken on August 24, 1992. (b) Template of the eye of the storm. (c) Correlation coefficient shown as an image (note the brightest point). (d) Location of the best match. This point is a single pixel, but its size was enlarged to make it easier to see. (Original image courtesy of NOAA.)

Reference

- Chapter 12 Feature extraction





Pattern Classification (chapter1) by R. O. Duda, P. E. Hart and D. G. Stork, John Wiley & Sons, 2000

Question

Thank you